

DIRECTED ENERGY WEAPONS: More On High-Energy Laser Weapons



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In this concluding part of the article, let us wrap up by learning some more about high-energy laser weapons.

Beam combination of multiple lasers

Power scale-up to ten to hundreds of kilowatts in a single laser, maintaining the desired beam quality, has many a technological challenges to overcome. Developments in the field of beam-combination techniques have opened new avenues of building higher-power lasers than could be combined in a single laser. Combining laser output from multiple fibre lasers is an important area of relevance to high-power directed energy laser weapon systems, as it allows achieving higher output power from relatively low-power individual lasers.

Direct diode lasers, bulk solid-state lasers and fibre lasers have been experimented with in recent years to generate higher powers by combining outputs of multiple lasers of a given type. Power enhancement and beam quality issues have been studied over propagation distances in kilometre range. Three common laser beam combining techniques include spectral beam combining, coherent beam combining and incoherent beam combining.

In the case of spectral beam combining, multiple laser beams with non-overlapping optical spectra are combined by using a

wavelength-sensitive beam combiner such as diffraction grating and prisms. Optical components with wavelength-sensitive transmission characteristics such as volume Bragg gratings and dichroic mirrors can also be used.

In the case of coherent beam combining, multiple lasers are combined to generate higher output powers with more or less the same beam quality as that of individual lasers. Coherent combining also preserves the spectral bandwidth.

In one method of incoherent beam combination, multiple laser beams are combined by overlapping individual laser beams on the target with a beam director consisting of independently-controlled steering mirrors. Adaptive optics may be used to compensate for the distortions caused by atmospheric turbulence. This technique is relatively much simpler than other beam-combining techniques including spectral and coherent beam combining. It does not require phase locking or polarisation locking of individual lasers and its power can be readily scaled up for directed energy weapon applications.

Representative directed energy laser weapon systems

Presently, a large number of directed energy laser weapon systems are reported to be under development and upgradation. Some of these are experimental, some technology demonstrators, while others are being upgraded and ruggedised to become realistic battlefield weapon systems in the near future.

Some of the more talked about laser weapon systems include high-energy laser system developed by Diehl and LFK of Germany, general-area defence integrated anti-missile laser system from TRW, mid infrared chemical laser from

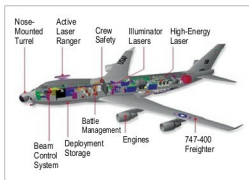


Fig. 10:
Airborne laser